

# Final Data Analytics Project Development

## Project Title

CoreTech Business Performance Dashboard

## Project Objective

The objective of this project is to analyze CoreTech Innovation's business data and develop an interactive dashboard that helps monitor business performance, client activity, project progress, revenue generation, and service effectiveness.

## Dataset Overview

The dataset contains 50 client records with information related to client names, company names, services, project budgets, project status, start and end dates, client cities, lead sources, payment status, and client ratings.

## DATASET

| A         | B           | C                       | D                      | E              | F              | G          | H         | I           | J           | K              | L             |
|-----------|-------------|-------------------------|------------------------|----------------|----------------|------------|-----------|-------------|-------------|----------------|---------------|
| Client_ID | Client_Name | Company_Name            | Service                | Project_Budget | Project_Status | Start_Date | End_Date  | Client_City | Lead_Source | Payment_Status | Client_Rating |
| C001      | Parkash     | FBR                     | Web Development        | 5000           | Completed      | 1/1/2026   | 1/3/2026  | Hyderabad   | Website     | Paid           | 4.5           |
| C002      | Sagar       | NovaTech Solutions      | Data Analytics         | 7000           | In Progress    | 3/1/2026   | 5/4/2026  | Karachi     | LinkedIn    | Paid           | 4.2           |
| C003      | Ajna        | Apex Innovations        | Mobile App Development | 6000           | Pending        | 5/1/2026   | 20/3/2026 | Islamabad   | Referral    | Pending        | 4.8           |
| C004      | Hina        | BrightEdge Technologies | SEO Services           | 2500           | In Progress    | 7/1/2026   | 15/4/2026 | Lahore      | Facebook    | Pending        | 4.1           |
| C005      | Ahsas       | CoreTech                | Graphic Design         | 3500           | Completed      | 10/1/2026  | 20/2/2026 | Hala        | Instagram   | Paid           | 4.6           |
| C006      | Shane Ali   | DigitalSphere Ltd       | UI/UX Design           | 8000           | In Progress    | 13/1/2026  | 25/4/2026 | Sukkur      | Website     | Pending        | 4.7           |
| C007      | Ruman       | NovaTech Solutions      | Graphic Design         | 5500           | Pending        | 15/1/2026  | 1/4/2026  | Nawabshah   | LinkedIn    | Paid           | 4.3           |
| C008      | Emaan       | CoreTech                | Web Development        | 4000           | In Progress    | 17/1/2026  | 5/5/2026  | Larkana     | Referral    | Pending        | 4             |
| C009      | Nade        | BrightEdge Technologies | UI/UX Design           | 4400           | Completed      | 20/1/2026  | 10/5/2026 | Hyderabad   | Facebook    | Paid           | 4.9           |
| C010      | Salar       | Apex Innovations        | Data Analytics         | 9000           | In Progress    | 23/1/2026  | 23/2/2026 | Karachi     | Instagram   | Pending        | 3.6           |
| C011      | Neelam      | BrightFuture Tech       | Web Development        | 6500           | Completed      | 25/1/2026  | 25/2/2026 | Islamabad   | Facebook    | Pending        | 3.9           |
| C012      | Nabeel      | BrightEdge Technologies | Data Analytics         | 8500           | In Progress    | 1/2/2026   | 10/5/2026 | Lahore      | Website     | Pending        | 4.1           |
| C013      | Isha        | NovaTech Solutions      | Mobile App Development | 5000           | Pending        | 3/2/2026   | 15/5/2026 | Hala        | LinkedIn    | Paid           | 4.3           |
| C014      | Nidesh      | Spark Digital           | SEO Services           | 3000           | In Progress    | 5/2/2026   | 20/4/2026 | Sukkur      | Referral    | Pending        | 4.2           |
| C015      | Harsha      | Apex Innovations        | Graphic Design         | 9200           | Completed      | 7/2/2026   | 25/5/2026 | Nawabshah   | Facebook    | Paid           | 5.5           |
| C016      | Saad        | CoreTech                | UI/UX Design           | 3500           | Pending        | 10/2/2026  | 18/4/2026 | Karachi     | Instagram   | Pending        | 4.8           |
| C017      | Sana        | SmartEdge Technologies  | Data Analytics         | 7000           | Completed      | 13/2/2026  | 27/4/2026 | Larkana     | Website     | Pending        | 4.9           |
| C018      | Laiba       | Vertex Digital          | Graphic Design         | 2500           | In Progress    | 15/2/2026  | 3/5/2026  | Nawabshah   | LinkedIn    | Pending        | 3.5           |
| C019      | Bital       | NovaTech Solutions      | Web Development        | 9700           | Pending        | 17/2/2026  | 10/5/2026 | Hyderabad   | Facebook    | Paid           | 3.7           |
| C020      | Maham       | BrightEdge Technologies | Web Development        | 3400           | In Progress    | 20/2/2026  | 17/5/2026 | Karachi     | Instagram   | Pending        | 3.8           |
| C021      | Iqra        | DigitalSphere Ltd       | Data Analytics         | 2700           | Completed      | 23/2/2026  | 20/6/2026 | Islamabad   | Facebook    | Paid           | 3.9           |

## Tools Used

- Microsoft Excel (Data Analysis & Cleaning)
- MySQL (SQL Queries)
- Power BI Desktop (Dashboard Development)

## Data Cleaning Process

The dataset was cleaned to improve data quality and accuracy. Missing values were identified and corrected, duplicate records were checked and removed, date formats were standardized, project budgets were formatted as currency, and text values were standardized for consistency.

## Missing Values Handling

|    | A         | B           | C                       | D                      | E              | F              | G          | H         | I           | J           | K              | L             |
|----|-----------|-------------|-------------------------|------------------------|----------------|----------------|------------|-----------|-------------|-------------|----------------|---------------|
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| 3  | C002      |             | NovaTech Solutions      | Data Analytics         | 7000           | In Progress    | 3/1/2026   | 5/4/2026  | Karachi     | LinkedIn    | Paid           | 4.2           |
| 4  | C003      | Ajna        | Apex Innovations        | Mobile App Development | 6000           | Pending        | 5/1/2026   | 20/3/2026 | Islamabad   | Referral    | Pending        | 4.8           |
| 5  | C004      | Hina        |                         | SEO Services           | 2500           | In Progress    | 7/1/2026   | 15/4/2026 | Lahore      | Facebook    | Pending        | 4.1           |
| 6  | C005      | Ahsas       | CoreTech                | Graphic Design         | 3500           | Completed      | 10/1/2026  | 20/2/2026 | Hala        | Instagram   | Paid           | 4.6           |
| 7  | C006      | Shane Ali   | DigitalSphere Ltd       | UI/UX Design           |                | In Progress    | 13/1/2026  | 25/4/2026 | Sukkur      | Website     | Pending        | 4.7           |
| 8  | C007      | Ruman       | NovaTech Solutions      | Graphic Design         | 5500           | Pending        | 15/1/2026  | 1/4/2026  | Nawabshah   | LinkedIn    | Paid           | 4.3           |
| 9  | C008      | Emaan       | CoreTech                | Web Development        | 4000           | In Progress    | 17/1/2026  | 5/5/2026  | Larkana     | Referral    | Pending        | 4             |
| 10 | C009      | Nade        | BrightEdge Technologies | UI/UX Design           | 4400           | Completed      | 20/1/2026  | 10/5/2026 | Hyderabad   | Facebook    | Paid           |               |
| 11 | C010      | Salar       | Apex Innovations        | Data Analytics         | 9000           | In Progress    | 23/1/2026  | 23/2/2026 | Karachi     | Instagram   | Pending        | 3.6           |
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| 13 | C012      | Nabeel      | BrightEdge Technologies | Data Analytics         | 8500           | In Progress    | 1/2/2026   | 10/5/2026 | Lahore      | Website     | Pending        | 4.1           |
| 14 | C013      | Isha        | NovaTech Solutions      | Mobile App Development | 5000           | Pending        | 3/2/2026   | 15/5/2026 | Hala        | LinkedIn    | Paid           | 4.3           |
| 15 | C014      | Nidesh      | Spark Digital           | SEO Services           | 3000           | In Progress    | 5/2/2026   | 20/4/2026 | Sukkur      | Referral    | Pending        | 4.2           |
| 16 | C015      | Harsha      | Apex Innovations        | Graphic Design         | 9200           | Completed      | 7/2/2026   | 25/5/2026 | Nawabshah   | Facebook    | Paid           | 5.5           |
| 17 | C016      | Saad        | CoreTech                | UI/UX Design           | 3500           | Pending        | 10/2/2026  | 18/4/2026 | Karachi     | Instagram   | Pending        | 4.8           |
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| 23 | C022      | Sheeraz     | BrightFuture Tech       | Mobile App Development | 8800           | In Progress    | 25/2/2026  | 23/5/2026 | Lahore      | Instagram   | Pending        | 4             |
| 24 | C023      | Hitik       | FBR                     | SEO Services           | 9500           | Completed      | 1/3/2026   | 5/5/2026  | Hala        | Referral    | Paid           | 4.9           |
| 25 | C024      | Disha       | SmartEdge Technologies  | Graphic Design         | 9000           | Pending        | 3/3/2026   | 10/5/2026 | Sukkur      | Website     | Pending        | 4.2           |
| 26 | C025      | Madhu       | Spark Digital           | UI/UX Design           | 4000           | In Progress    | 5/3/2026   | 15/5/2026 | Nawabshah   | LinkedIn    | Paid           | 4.8           |

## Duplicate Records Check

| 1  | Client_ID | Client_Name | Company_Name            | Service                | Project_Budget | Project_Status | Start_Date | End_Date  | Client_City | Lead_Source | Payment_Status | Client_Rating |
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| 24 | C024      | Disha       | SmartEdge Technologies  | Graphic Design         | 9000           | Pending        | 3/3/2026   | 10/5/2026 | Sukkur      | Website     | Pending        | 4.2           |

## Pivot Table Analysis

Several pivot tables were created to summarize business information and identify patterns.

Pivot tables included:

- Total Budget by Service

- Number of Clients by City
- Project Status Count
- Lead Source Performance
- Payment Status Summary
- Average Rating by Service

These pivot tables helped identify top-performing services, client distribution, project progress, and customer satisfaction levels.

## PIVOT TABLES

| project status count |                         |
|----------------------|-------------------------|
| Row Label            | Count of Project_Status |
| Completed            | 16                      |
| In Progress          | 20                      |
| Pending              | 14                      |
| Grand Total          | 50                      |

| payment status summary |                         |
|------------------------|-------------------------|
| Row Label              | Count of Payment_Status |
| Paid                   | 24                      |
| Pending                | 26                      |
| Grand Total            | 50                      |

| lead source performance |                      |
|-------------------------|----------------------|
| Row Label               | Count of Lead_Source |
| Facebook                | 10                   |
| Instagram               | 9                    |
| LinkedIn                | 12                   |
| Referral                | 10                   |
| Website                 | 9                    |
| Grand Total             | 50                   |

| total budget by service |                       |
|-------------------------|-----------------------|
| Row Labels              | Sum of Project_Budget |
| Data Analytics          | \$62,500.00           |
| Graphic Design          | \$61,600.00           |
| Mobile App Development  | \$42,000.00           |
| SEO Services            | \$34,000.00           |
| UI/UX Design            | \$45,100.00           |
| Web Development         | \$52,700.00           |
| Grand Total             | \$297,900.00          |

| average rating by service |                      |
|---------------------------|----------------------|
| Row Labels                | Sum of Client_Rating |
| Data Analytics            | 41                   |
| Graphic Design            | 48.5                 |
| Mobile App Development    | 26.3                 |
| SEO Services              | 21.8                 |
| UI/UX Design              | 39.7                 |
| Web Development           | 37.1                 |
| Grand Total               | 214.4                |

The pivot tables summarize business data and help identify trends in services, cities, payments, and ratings.

## Charts and Data Visualization

Different charts were created to visually analyze business performance.

Charts created:

- Revenue by Service (Bar Chart)
- Project Status Distribution (Pie Chart)
- Clients by City (Column Chart)

- Projects by Date (Line Chart)
- Payment Status (Donut Chart)
- Budget and Rating by Service (Combo Chart)

These visualizations made it easier to identify trends and compare performance across different categories.

## CHARTS



The charts highlight revenue trends, project status distribution, client locations, and payment performance.

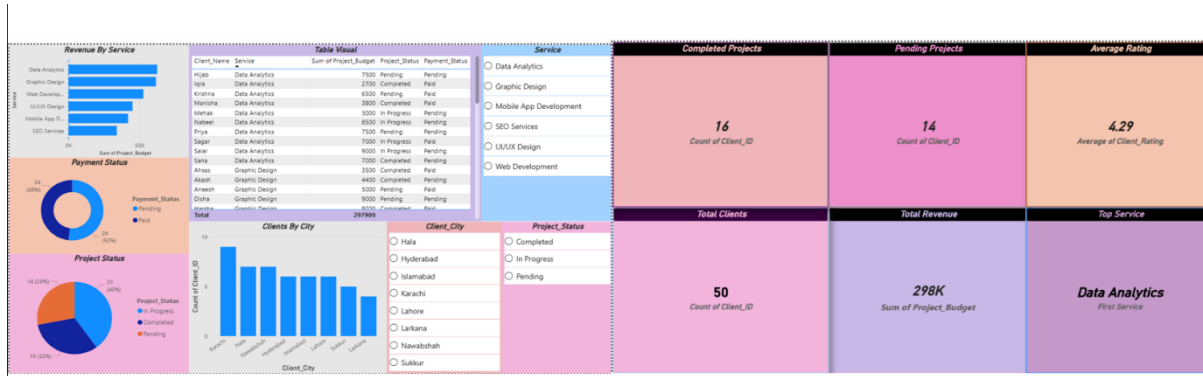
## Power BI Dashboard

A business dashboard was developed in Power BI to provide an overview of company performance through interactive visualizations and KPIs.

Dashboard Features:

- Revenue Analysis
- Project Status Analysis
- Client Distribution
- Payment Status Tracking
- Service Performance Analysis

## DASHBOARD



This dashboard provides a visual overview of business performance, project distribution, payment status, and client analysis.

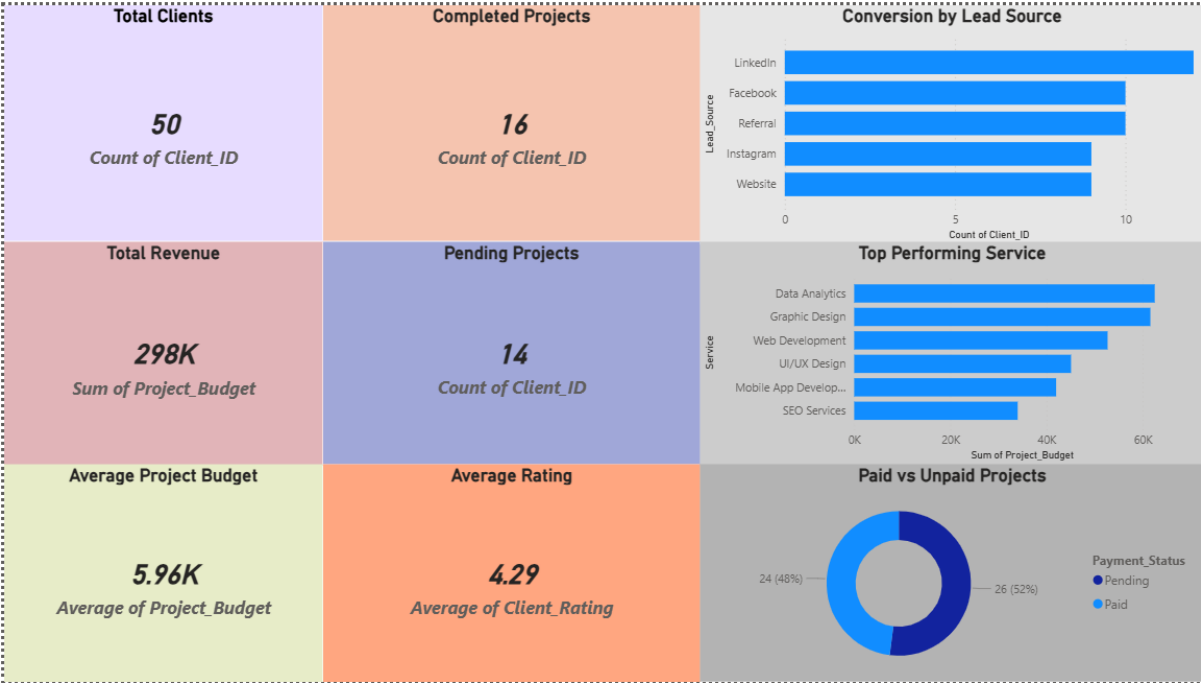
## KPI Dashboard

A KPI dashboard was created to track key business metrics.

KPIs Included:

- Total Clients
- Total Revenue
- Average Project Budget
- Completed Projects
- Pending Projects
- Average Rating
- Top Performing Service
- Best Lead Source
- Paid vs Unpaid Projects

KPI DASHBOARD



This dashboard presents key business metrics including total revenue, total clients, project status, average rating, and service performance.

Business Insights

1. CoreTech Innovation has successfully served 50 clients.
2. Total revenue generated is approximately 298K.
3. Data Analytics is the highest revenue generating service.
4. LinkedIn is the most effective lead source.
5. The average project budget is approximately 5.96K.

6. The average client rating is 4.29, indicating good customer satisfaction.
7. Several projects are still pending and require closer monitoring.
8. Some project payments remain unpaid, which may impact cash flow management.

## **Recommendations**

1. Invest more resources in Data Analytics services.
2. Increase marketing activities on LinkedIn.
3. Improve project tracking and monitoring processes.
4. Strengthen payment collection procedures.
5. Promote lower-performing services through targeted marketing campaigns.

## **Conclusion**

This project successfully analyzed CoreTech Innovation's business performance using Excel, MySQL, and Power BI. The dashboard provides valuable insights into revenue, project status, client acquisition, payment performance, and service effectiveness. The findings can support better business decisions and improve future growth opportunities.